7B Circumference and Area of Circles and Sectors

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. Find the exact circumference C (in terms of π) of each circle:

(a) Radius $r = 7\ \text{cm}$ (b) Diameter $d = 18\ \text{m}$ (c) Radius $r = 4.5\ \text{cm}$ (d) Diameter $d = 25\ \text{mm}$

Q2. Find the exact area A (in terms of π) of each circle:

(a) Radius $r = 6\ \text{cm}$ (b) Radius $r = 11\ \text{m}$ (c) Diameter $d = 16\ \text{cm}$ (d) Diameter $d = 30\ \text{m}$

Q3. Find the exact area and exact total perimeter (including straight edges) of each sector:

(a) Semicircle of radius $r = 10\ \text{cm}$ (b) Quadrant of radius $r = 6\ \text{m}$ (c) Sector with $\theta = 60^\circ, r = 9\ \text{cm}$ (d) Sector with $\theta = 120^\circ, r = 12\ \text{cm}$

Q4. Using $\pi \approx \dfrac{22}{7}$, evaluate each circular measurement:

(a) Circumference of a circle with $r = 21\ \text{cm}$ (b) Area of a circle with $r = 7\ \text{m}$ (c) Perimeter of a semicircle with $d = 14\ \text{cm}$ (d) Area of a quadrant with $r = 14\ \text{cm}$

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. A circular cricket oval in Melbourne has a boundary radius of $70\ \text{m}$. Using $\pi \approx \dfrac{22}{7}$, find (a) the length of the boundary fence around the oval and (b) the turf area of the oval in square metres and in hectares.

Q6. A circular pond of radius $4\ \text{m}$ is surrounded by a gravel path of width $2\ \text{m}$ (forming an outer radius of $6\ \text{m}$). Find the exact area of the gravel path in terms of π, and round your answer to 2 decimal places.

Q7. A windscreen wiper blade of length $45\ \text{cm}$ sweeps through an angle of 120° from its pivot. Find the exact area of glass cleaned in a single sweep in terms of π.

Q8. A rectangular sheet of steel measuring $20\ \text{cm}$ by $10\ \text{cm}$ has two identical largest-possible circles of diameter $10\ \text{cm}$ ($r = 5\ \text{cm}$) stamped out of it. Find the exact area of steel remaining.

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. A piece of wire of length $44\ \text{cm}$ is bent into a square, and an identical $44\ \text{cm}$ wire is bent into a circle. Using $\pi \approx \dfrac{22}{7}$, find the area of both shapes and determine which shape encloses the larger area.

Q10. If the circumference of a circle is $16\pi\ \text{cm}$, show how to work backwards to find its radius r and then its exact area A.